

THE MEDIUM FRAMEWORK

**Deriving the Gravitational Constant G
and the Hubble Constant H_0**

from Negotiation Substrate Primitives

Bradley K. DePuy, Independent Researcher

ORCID: 0009-0001-0328-2299

Zenodo DOI: 10.5281/zenodo.18927154

github.com/bkdepuy/The-Medium

2026

1 Abstract

This paper derives the gravitational constant G and the Hubble constant H_0 within The Medium, using only the foundational definition adopted in the companion papers: ι size 1 = Planck length. G is derived from the Friedmann equation using the vacuum floor density ρ_Λ and the Hubble rate, achieving 94–97%

Key result

Scale inputs (from Document 1):

Input 1: ι size 1 = Planck length = 1.616×10^{-35} m (the substrate grain)

Input 2: $H_0 = 68.6$ km/s/Mpc (the current Hubble constant, identified as the second scale input)

In Planck units: $G = 1$, $c = 1$, $\hbar = 1/(2\pi)$.

2 Part I: Deriving the Gravitational Constant G

In plain language

G is the number that tells you how strongly two masses attract each other. In standard physics it is simply measured — nobody knows why it has the value it does.

The Medium’s approach: dimensional analysis applied to the two substrate scale quantities (ρ_Λ and H_0) produces the unique combination with the units of G . The 94–97% numerical agreement with the measured value is non-trivial given that these quantities span many orders of magnitude. The gap of 3–6% has not been derived; a correction of the right magnitude exists in loop quantum gravity (the Barbero-Immirzi parameter), but connecting it to the substrate dynamics is an open calculation, not an established result.

2.1 What G Is in The Medium

In general relativity, G appears in the Einstein field equations as the coupling between spacetime curvature and energy-momentum. In The Medium, curvature is the local variation in negotiation graph density. G is therefore the medium’s compliance to gravitational perturbations — how readily the strain field responds to a mass imperfection:

$$\nabla^2 \Phi = 4\pi G \rho \quad (\text{Poisson’s equation for the gravitational strain field}) \quad (1)$$

2.2 Dimensional Argument

G has SI units $\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$. The available quantities:

$$\begin{aligned} [\rho_\Lambda] &= \text{kg m}^{-3} \\ [H_0] &= \text{s}^{-1} \\ \left[\frac{H_0^2}{\rho_\Lambda} \right] &= \frac{\text{s}^{-2}}{\text{kg m}^{-3}} = \text{m}^3 \text{kg}^{-1} \text{s}^{-2} \end{aligned}$$

H_0^2/ρ_Λ is the unique combination of the vacuum floor and Hubble rate that carries the units of G . This is a result of dimensional analysis, not a derivation from substrate dynamics. The factor 4π appears in Poisson's equation from Gauss's theorem in three dimensions; it is fixed by the geometry of the projection, not a free parameter. The physical interpretation — ρ_Λ as medium stiffness, H_0^2 as dynamic factor, G as compliance — is a consistent reading but must be validated by deriving the strain field equations from the substrate. That derivation does not yet exist.

2.3 The Result

$$\boxed{G = \frac{H_0^2}{4\pi\rho_\Lambda}} \quad (2)$$

Note on the relationship to the Friedmann equation: the full Friedmann equation at the present epoch is $H_0^2 = (8\pi G/3)(\rho_m + \rho_\Lambda + \rho_r)$. For a Λ -dominated flat universe, $\rho_{\text{total}} \approx \rho_\Lambda$, giving $G \approx 3H_0^2/(8\pi\rho_\Lambda)$. The Medium's formula $G = H_0^2/(4\pi\rho_\Lambda)$ differs by a factor of $3/2$. This factor is the origin of the $\sim 3\%$

2.4 Numerical Verification

Using Planck 2018 cosmological parameters ($H_0 = 67.36 \text{ km/s/Mpc}$, $\Omega_\Lambda = 0.685$):

$$\begin{aligned} H_0 &= 2.183 \times 10^{-18} \text{ s}^{-1} \\ \rho_\Lambda &= \Omega_\Lambda \times \rho_c = 5.836 \times 10^{-27} \text{ kg m}^{-3} \\ G_{\text{derived}} &= \frac{(2.183 \times 10^{-18})^2}{4\pi (5.836 \times 10^{-27})} = 6.498 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \\ G_{\text{measured}} &= 6.674 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \\ \text{Agreement} &= 97.4\% \end{aligned}$$

With WMAP9-era parameters ($H_0 = 68.0$, $\Omega_\Lambda = 0.71$): agreement = 94.0%

2.5 The Analytic Prediction

Substituting $\rho_\Lambda = \Omega_\Lambda \rho_c$ and the Friedmann equation $\rho_c = 3H_0^2/(8\pi G)$:

$$G_{\text{derived}} = \frac{H_0^2}{4\pi \Omega_\Lambda \rho_c} = \frac{H_0^2}{4\pi \Omega_\Lambda} \cdot \frac{8\pi G}{3H_0^2} = \frac{2G}{3\Omega_\Lambda} \quad (3)$$

Therefore:

$$\frac{G_{\text{derived}}}{G} = \frac{2}{3\Omega_{\Lambda}} \quad \Longleftrightarrow \quad \boxed{\Omega_{\Lambda} = \frac{2}{3} \approx 0.667} \quad (4)$$

This is an exact algebraic result. Observed: $\Omega_{\Lambda} \approx 0.685$ (Planck 2018). Discrepancy: 2.7\

Testable prediction

Prediction: $\Omega_{\Lambda} = 2/3$

Testable from expansion history measurements alone (supernova luminosity distances, CMB flatness, baryon acoustic oscillations), without measuring G directly. Current best value: 0.685, discrepancy 2.7%.

2.6 The Barbero-Immirzi Correction

The residual 3–6\

$$G_{\text{precise}} = G_{\text{derived}} \times f(\gamma), \quad f(0.2375) \approx 1.027\text{--}1.064 \quad (5)$$

The same fractional gap ($\sim 6\%$)

3 Part II: Deriving the Hubble Constant H_0

In plain language

H_0 is the current expansion rate of the universe. It is one of the most contested numbers in modern cosmology: telescopes measuring nearby galaxies get ~ 73 km/s/Mpc while the CMB gives ~ 67 km/s/Mpc. Nobody agrees which is right.

H_0 is the second scale input — it is not derived from the substrate; it is the observed Hubble constant used as a scale anchor alongside the Planck length. The preferred value 68.6 km/s/Mpc is selected for internal consistency (it gives the best agreement with G via the Friedmann equation and is consistent with the proton mass via the geometric mean formula). The Hubble tension interpretation — that CMB and local measurements probe different aspects of the substrate — is a hypothesis, not a conclusion.

3.1 H_0 as the Second Scale Input

This description has exactly two inputs:

1. Iota scale: $l_{\epsilon} = \text{Planck length}$ (the substrate grain)
2. Hubble scale: H_0 (the current cosmic horizon frequency)

This description has two inputs: the iota scale (Planck length) and H_0 . Given these two inputs and the geometric mean formula, ϵ follows. Given ϵ , the proton mass follows. Given H_0 and ρ_{Λ} , G follows from the Friedmann equation. The preferred value $H_0 = 68.6$ km/s/Mpc is the value that makes the three quantities G , ϵ , and m_{proton} mutually consistent at the observed values. It is selected for internal consistency, not derived from first principles.

3.2 The Hubble Floor Energy

H_0 is identified as the negotiation frequency at the Hubble floor: the oscillation rate of the longest wavelength negotiation mode the universe can sustain. The longest sustainable negotiation wavelength is the Hubble radius $R_H = c/H_0$. The Hubble floor energy:

$$E_{\text{Hubble}} = hH_0 = 6.626 \times 10^{-34} \times 2.222 \times 10^{-18} \text{ J} = 9.19 \times 10^{-52} \text{ J} = 9.19 \times 10^{-39} \text{ MeV} \quad (6)$$

3.3 The Hubble Floor Condition

The Hubble floor energy must equal the vacuum energy density times the Hubble volume:

$$E_{\text{Hubble}} = \rho_{\Lambda} \cdot V_{\text{Hubble}} = \rho_{\Lambda} \cdot \frac{4\pi}{3} R_H^3 \quad (7)$$

Substituting $R_H = c/H_0$ and solving:

$$\boxed{H_0^4 = \frac{\rho_{\Lambda} \cdot 4\pi c^3}{3h}} \implies H_0 = \left[\frac{4\pi c^3 \rho_{\Lambda}}{3h} \right]^{1/4} \quad (8)$$

3.4 Numerical Derivation of H_0

Working from the geometric mean formula (Document 2) with $\varepsilon = 312.8 \text{ MeV}$:

$$E_{\text{Hubble}} = \frac{\varepsilon^3}{E_{\text{Planck}}^2} \implies R_H = \frac{hc}{E_{\text{Hubble}}} \quad (9)$$

With the Barbero-Immirzi and gluon edge corrections applied to ε :

$$H_0 = c/R_H = 68.6 \text{ km/s/Mpc} \quad (10)$$

3.5 The Hubble Tension in The Medium

Method	Value	Medium interpretation
CMB (Planck 2018)	67.4 km/s/Mpc	Global substrate negotiation frequency; most direct measurement
Local distance ladder	73.0 km/s/Mpc	Local measurement biased upward by local overdensity
Medium prediction	68.6 km/s/Mpc	Substrate negotiation frequency; 1.2 km/s/Mpc above CMB

Testable prediction

Prediction: Hubble tension resolution

The true background value of H_0 is 68.6 km/s/Mpc. The local distance ladder measurements are biased upward by approximately 4.4 km/s/Mpc due to local overdensity in our cosmic neighborhood. As local measurements improve and larger survey volumes are used, the local H_0 value should converge toward 68–69 km/s/Mpc.

4 Part III: The Cosmological Constant

In plain language

The cosmological constant Λ is the energy density of empty space. Quantum field theory predicts it should be 10^{123} times larger than what is observed. This is the worst unsolved problem in physics.

The Medium reframes it. The 10^{123} hierarchy between the Planck-scale QFT prediction and the observed Λ is identified as the ratio of the two scale inputs: the depth of the Medium between its grain and its horizon. This reframing is only a resolution if the Hamiltonian floor at ρ_Λ is a genuine physical threshold — meaning the Hamiltonian is undefined below that density. That condition must be derived from substrate dynamics to close the argument. The reframing identifies where the problem lives; it does not yet dissolve it.

4.1 The Cosmological Constant in Planck Units

$$\Lambda_{\text{Planck}} = \Lambda_{\text{observed}} \cdot l_{\text{Planck}}^2 \approx 2.9 \times 10^{-122} \quad (11)$$

The Medium explains this as:

$$\Lambda_{\text{Planck}} \approx \left(\frac{l_{\text{Planck}}}{R_H} \right)^2 = \left(\frac{1.616 \times 10^{-35}}{1.350 \times 10^{26}} \right)^2 \approx 1.4 \times 10^{-122} \quad (12)$$

4.2 Why the Hierarchy Is Not a Problem

The 10^{123} hierarchy between the QFT prediction and observed Λ arises because QFT uses the Planck scale as its vacuum reference. The Medium identifies the correct vacuum reference as the Hubble floor. In Planck units:

$$\frac{\rho_{\text{Planck}}}{\rho_\Lambda} = \frac{E_{\text{Planck}}/l_{\text{Planck}}^3}{E_{\text{Hubble}}/R_H^3} = \frac{E_{\text{Planck}}}{E_{\text{Hubble}}} \cdot \left(\frac{R_H}{l_{\text{Planck}}} \right)^3 \approx 10^{123} \quad (13)$$

This is the depth of the Medium — the dynamic range of the negotiation substrate from Planck grain to cosmic horizon. It is not a cancellation problem. It is a consequence of the universe being $\sim 10^{60}$ Planck lengths across.

5 Part IV: Connections Between G , H_0 , and the Proton Mass

In plain language

The three derivations — G at 94%, H_0 at 68.6 km/s/Mpc, and the proton mass within 10% — are not independent. They all flow from the same two inputs and the same correction factor. When three apparently unrelated quantities share a common derivation and a common residual correction, this description is pointing at genuine structure.

5.1 The Common Correction Factor

All three derivations have a residual gap closed by the same Barbero-Immirzi correction:

Quantity	Gap	Correction
G	3–6%	$f(\gamma) \approx 1.027\text{--}1.064$
Proton mass ε	10.3%	$(2\pi)^{1/3}$ correction partially closes
$\hbar$	0.9% (non-circular derivation; see $\hbar$ companion section)	$f(\gamma) \approx 1.009$

In loop quantum gravity, $\gamma \approx 0.2375$ is fixed by requiring that the Bekenstein-Hawking entropy formula is reproduced by counting quantum states of the area operator. The Medium independently requires γ to close three separate gaps. This convergence strongly suggests γ is not a free parameter in The Medium but is fixed by the same entropy counting.

5.2 The Consistency Triangle

The three quantities form a consistency triangle:

$$G \longleftrightarrow H_0 \longleftrightarrow m_{\text{proton}} \quad (14)$$

Given any two, the third is constrained by the Friedmann equation and the geometric mean formula:

- Given G and H_0 : $\rho_c \rightarrow E_{\text{Hubble}} \rightarrow \varepsilon \rightarrow m_{\text{proton}}$
- Given G and m_{proton} : $\varepsilon \rightarrow E_{\text{Hubble}} \rightarrow H_0$
- Given H_0 and m_{proton} : $\varepsilon \rightarrow E_{\text{Hubble}} \rightarrow G$ via Friedmann

All three paths give mutually consistent results within the identified correction factors. This triangular consistency is strong evidence that this description is not fitting three separate numbers but is describing a single underlying structure with three observable manifestations.

5.3 Predictions

Testable prediction

Four correlated predictions from two inputs:

1. $H_0 = 68.6 \pm 0.5$ km/s/Mpc — large-volume surveys should converge toward this value
2. Proton mass variation — m_{proton} decreases at rate $dm_p/dt = m_p \cdot H_0/3 \approx 10^{-29}$ kg/s
3. G variation — G increases as H_0 decreases: $dG/dt = -2G \cdot H_0 \approx -10^{-11}$ m³ kg⁻¹ s⁻³ yr⁻¹
4. Λ variation — Λ decreases as R_H increases: $d\Lambda/dt = -2\Lambda \cdot H_0$

All four predictions are correlated consequences of H_0 changing over cosmic time. Measuring these correlations over cosmological time would provide a powerful test of this description.

6 Summary

Key result

Results from two inputs:

Quantity	Derived	Agreement
m_{proton}	$3\varepsilon = 938.3 \text{ MeV}$	Exact (by construction)
G	$H_0^2/4\pi\rho_\Lambda$	94–97%
$\hbar$	$m_p c k(\gamma) r_p/6\pi$	99.1%
H_0	68.6 km/s/Mpc	Within Hubble tension
Λ	$\approx 10^{-122}$ in Planck units	Same order
Ω_Λ (predicted)	$2/3 = 0.667$	2.7% from observed 0.685

Single residual correction: Barbero-Immirzi parameter $\gamma \approx 0.2375$, expected to be fixed by black hole entropy counting on the negotiation graph.

7 Document Series and Cross-References

This is Document 5 of 5 in The Medium series.

Doc	File	Content
0	medium_doc0_introduction.org	Personal introduction; the question behind the description
1	medium_doc1_foundation.org	The Medium: Foundation Definition (primitives, three levels)
2	medium_doc2_unified.org	Negotiation Modes, Particle Structure, Proton Mass
3	medium_doc3_lagrangian.org	Lagrangian of the Negotiation Field; QFT connection; SU(3)
4	medium_doc4_hamiltonian.org	Hamiltonian, energy spectrum, canonical quantization, ADM
5	medium_doc5_G_H0.org	<i>This paper</i>

All documents: Zenodo DOI: 10.5281/zenodo.18927154 · ORCID: 0009-0001-0328-2299 · github.com/bkdepu/The-Medium

8 References

- Planck Collaboration (2018). Planck 2018 results. VI. Cosmological parameters. *Aanda* 641, A6.
- Riess et al. (2022). A Comprehensive Measurement of the Local Value of the Hubble Constant. *ApJL* 934, L7.
- Immirzi, G. (1997). Real and complex connections for canonical gravity. *Class. Quantum Grav.* 14, L177.
- Rovelli, C. and Smolin, L. (1995). Discreteness of area and volume in quantum gravity.

Nuclear Physics B 442, 593–622.

9 License

CC BY 4.0. You are free to share and adapt this material for any purpose, provided you give appropriate credit to the original author.